

Donghyun Son

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INTRODUCTION

I'm an incoming Master's student in Computer Science at Stanford, working at the intersection of ML and Systems. I'm looking for research or engineering internship opportunities for 2027 summer.

EDUCATION

Stanford University , <i>M.S. in Computer Science (Incoming)</i>	Sep 2026 (expected) —
Seoul National University , <i>B.S. in Computer Science</i> GPA: 4.04/4.3 (4.11/4.3 in major), Summa Cum Laude	Mar 2018 — Feb 2026
University of Texas at Austin , <i>Exchange Student, Computer Science</i>	Sep 2025 — Dec 2025

PUBLICATIONS

*: equal contribution.

[W2] **EntropyCache: Decoded Token Entropy Guided KV Caching for Diffusion Language Models**

Minsoo Cheong, **Donghyun Son**, Woosang Lim, Sungjoo Yoo

AdaptFM: Resource-Adaptive Foundation Model Inference @ ICML 2026 (**AdaptFM @ ICML 2026**)

[P1] **NALAR: A Serving Framework for Agent Workflows**

Benedict Marco Laju, **Donghyun Son**, Saurabh Agarwal, Nitin Kedia, Myungjin Lee, Jayanth Srinivasa, Aditya Akella
Preprint, Under Review

[C2] **NSNQuant: A Double Normalization Approach for Calibration-Free Low-Bit Vector Quantization of KV Cache**

Donghyun Son, Euntae Choi, Sungjoo Yoo

Advances in Neural Information Processing Systems (**NeurIPS 2025**)

[W1] **Gradient Estimation for Unseen Domain Risk Minimization with Pre-Trained Models**

Byounggyu Lew*, **Donghyun Son***, Buru Chang

Workshop and Challenges for Out-of-Distribution Generalization in Computer Vision @ ICCV 2023 (**OOD-CV@ICCV 2023**)

[J1] **Looking to Personalize Gaze Estimation Using Transformers**

Seung Hoon Choi, **Donghyun Son**, Yunjong Ha, Seonghun Hong, Taejung Park

Journal of Computing Science and Engineering (**JCSE**), Vol. 17, No. 2, pp.41-50

[C1] **Reliable Decision from Multiple Subtasks through Threshold Optimization: Content Moderation in the Wild**

Donghyun Son*, Byounggyu Lew*, Kwanghee Choi*, Yongsu Baek, Seungwoo Choi, Beomjun Shin, Sungjoo Ha, Buru Chang

ACM International Conference on Web Search and Data Mining (**WSDM 2023, Oral**)

RESEARCH EXPERIENCE

AI Scientist Intern, EXAONE Lab LG AI Research	Apr 2026 — Present Seoul, South Korea
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- Optimizing inference pipelines of EXAONE models, one of the largest model families (> 200B) in South Korea

Undergraduate Researcher UTNS Research Group @ UT Austin Supervised by Prof. Aditya Akella	Aug 2025 — Dec 2025 Austin, Texas
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- Designed and implemented an agentic system where the central controller dynamically manages resource allocation, load balancing, and scheduling based on a global observability (submitted to NSDI26)

Undergraduate Research Intern CMALab @ SNU Supervised by Prof. Sungjoo Yoo	Sep 2024 — Aug 2025 Seoul, South Korea
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- Investigated a calibration-free vector quantization method for KV cache, incorporating a novel normalization algorithm and efficient CUDA kernels for 1–2 bit inference (NeurIPS25)

WORK EXPERIENCE

- Machine Learning Engineer, Moderation ML Team

Hyperconnect (Acquired by Match Group)

Mar 2022 — Jul 2023

Seoul, South Korea

 - Built an ML-based system for moderating images and audio across Match Group brands (e.g., Tinder, Hinge)
 - Devised a threshold optimization algorithm based on GD to optimize the tradeoff between reliability and cost (WSDM23, Oral)
 - Proposed a training algorithm for ML models to generalize effectively across different services and domains (OOD-CV@ICCV23)
- Research Engineer, R&D Team

VisualCamp

Aug 2020 — Mar 2022

Seoul, South Korea

 - Designed and implemented C++-based multi-threaded ML inference pipeline for low-latency gaze estimation
 - Built a robust training procedure to train an appearance-based gaze estimation model and improved accuracy by 30%
 - Proposed a transformer-based calibration algorithm that predicts user-specific latent vectors using users' samples

ACADEMIC SERVICES

- Conference Reviewer: NeurIPS 2026
- Workshop Reviewer: AdaptFM@ICML 2026

PATENTS

- Apparatus and Method for Setting Criteria on Data Classification (US20240281494A1)*
Yong Su Baek, **Dong Hyun SON**, Beom Jun Shin, Byoung Gyu Lew, Bu Ru CHANG, Kwang Hee CHOI, Seung Woo Choi, Sung Joo Ha
- Apparatus for Domain Generalization of Machine Learning Models, Methods and Computer Readable Recording Mediums Therefor (US20240087294A1)*
Bu Ru CHANG, Byoung Gyu Lew, **Dong Hyun SON**

HONORS & AWARDS

- Programming Contests**
- 2nd prize, at Union of Clubs Programming Contest (UCPC) 2025
 - 5th prize, at Samsung Collegiate Programming Contest (SCPC) 2021
 - 5th place, at ICPC NERC Huawei Challenge 2020: Cloud Scheduling Challenge
 - 1st place, at Seoul National University Programming Contest (SNUPC) 2019, div. 2
- Scholarships**
- Samsung Software Membership, from Samsung Research (Nov 2021 — Feb 2026)
 - Korea-U.S. STEM Student Exchange Scholarship (\$9000), from Ministry of Trade, Industry and Energy (2025.09)
 - Merit-based Full Tuition Academic Scholarship (\$2300), from Seoul National University (2020-1, 2023-2, 2024-1, 2024-2)
- Others**
- 4th place, at CXR-LT Challenge Task 1 (MICCAI 2024)
 - Bronze Prize, at 42nd Undergraduate Mathematics Contest div.1 (for mathematical majors)
 - Microsoft Azure Champ Prize : Hack For Good, at TartanHacks 2021 in CMU

PRESENTATIONS

- KV Cache Compression for Long Context Inference, Weekly Seminar @ Deepest S16 (January 2025)
- Efficient Algorithms for LLM Inference, Weekly Seminar @ Deepest S15 (July 2024)

ADDITIONAL INFORMATION

- References:** Prof. Sungjoo Yoo (Seoul National University), Prof. Buru Chang (Korea University), Dr. Saurabh Agarwal (UT Austin)
- Interests:** Table Tennis, Math Puzzles, Mind Sports (4 Dan in Go)
- Technical Skills:** C++, Python, Pytorch, Tensorflow, Jax, CUDA, OpenCL, MPI, OpenMP
- Problem Solving:** Codeforces (handle: **diordhd**, rating: 2200+) / BOJ (handle: **dhdroid**)
- TOEFL iBT:** 108 (R29/L29/S22/W28)